some results are statistically significant, we are unable to determine clinical significance with our limited data. Another limitation is that Area Health Resources File data uses a single mailing address per provider; thus, we were unable to account for providers who may practice in multiple locations or travel across county lines to work in satellite clinics. In addition, the distance between neighboring counties with and those without dermatologists was not calculated. It is possible that patients in counties without a dermatologist could have had access to a provider in a nearby county. Future work could further explore the correlations between patient outcomes and dermatologist density and the effects of telemedicine on patient access to care.

DATA AVAILABILITY STATEMENT

Data are public data from the Area Health Resources File and can be requested (https://data.hrsa.gov/topics/health-workforce/ahrf).

KEYWORDS

Patient access; Poverty; Practice patterns; Rurality; Workforce trends over time

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CONFLICT OF INTEREST

The authors state no conflict of interest.

ACKNOWLEDGMENTS

Institutional review board approval was not required by Stanford University because the study used publicly available data and did not involve human subjects.

AUTHOR CONTRIBUTIONS

Conceptualization: HG, GHB; Data Curation: HG; Formal Analysis: HG; Supervision: GHB; Visualization: HG; Writing - Original Draft Preparation: HG, GHB; Writing - Review and Editing: HG, GHB

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The Dual Roles of Leucine-Rich Repeat-Containing Protein 15 Positive Fibroblasts: From Cancer to Tissue Repair

Journal of Investigative Dermatology (2025) **145,** 200–204; doi:10.1016/j.jid.2024.06.1279

TO THE EDITOR

Growing consensus suggests that leucine-rich repeat-containing protein 15 positive (LRRC15+) fibroblasts are cancer-associated fibroblasts and play crucial roles in tumor progression. Krishnamurty et al (2022) have presented compelling evidence that LRRC15 expression is confined to tumor-associated fibroblasts. Targeted depletion of LRRC15+ cancer-associated fibroblasts reduces the overall tumor fibroblast content, amplifies the effector function of a tumor-infiltrating CD8+ T cells, and enhances tumor regression in response to anti-programmed death-ligand 1 immune checkpoint blockade. This study suggests that selective depletion of LRRC15+ cancer-associated fibroblasts might offer potential as a new therapeutic strategy, leading to robust and enduring responses to cancer immunotherapy. However, LRRC15+ fibroblasts are not exclusive to the cancer microenvironment. They have also been identified in conditions such as arthritis, skin wounds, and lung fibrosis in mice, as well as in human tissues affected by lung fibrosis or ulcerative colitis (Buechler et al, 2021). Moreover, LRRC15 protein expression has been detected in several human tissues under physiological conditions, including hair follicles, tonsils, certain regions of the stomach (cardia and pylorus), the spleen (peritrabecular region), osteoblasts in the bone, and areas undergoing wound healing (Purcell et al, 2018). LRRC15 mRNA has also been detected in hair follicles and the

Abbreviations: ECM, extracellular matrix; LRRC15+, leucine-rich repeat-containing protein 15 positive

Accepted manuscript published online 8 July 2024; corrected proof published online 29 July 2024

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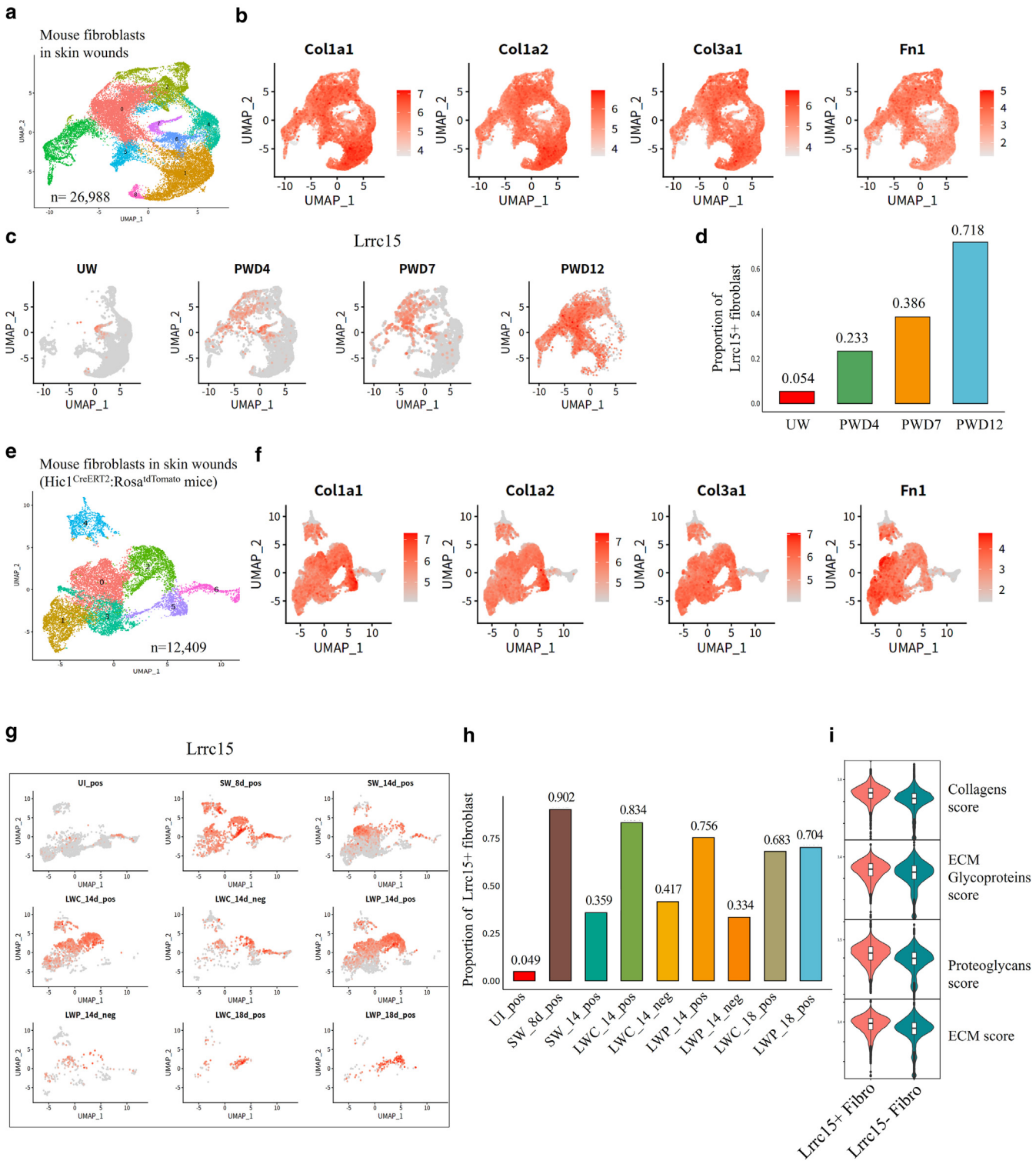

Figure 1. LRRC15+ fibroblast dynamics during mouse skin wound healing. (**a**) UMAP visualization capturing 26,988 fibroblasts at various time points during wound healing in mice, including the samples from uninjured skin and from 4, 7, and 12 days after wounding. (**b**) UMAP plot color-coded based on the expression of fibroblast marker genes COL1A1, COL1A2, COL3A1, and FN1. (**c**) UMAP plot color-coded according to the LRRC15 expression in the UW, PWD4, PWD7, and PWD12 groups. (**d**) Percentage of LRRC15+ fibroblasts at different time points. (**e**) UMAP visualization of 12,409 fibroblasts at different time points in Hic1^{CreERT2}:Rosa^{tdTomato} mice during wound healing, included uninjured skin and 8, 14, and 18 days after wounding. (**f**) UMAP plot color-coded based on the expression of fibroblast marker genes COL1A1, COL1A2, COL3A1, and FN1. (**g**) The UMAP plot is color-coded according to the LRRC15 expression in different groups. Following tamoxifen treatment in Hic1^{CreERT2}:Rosa^{tdTomato} mice, Hic1+ and Hic1- cells were then isolated from various areas using FACS. (**h**) The fraction of LRRC15+ fibroblasts is analyzed at different groups. (**i**) Violin plots present scores for Collagens, ECM glycoproteins, proteoglycans, and total ECM in human LRRC15+ and LRRC15- fibroblast clusters. COL1A1, collagen type I Alpha 1 Chain; COL1A2, collagen type I alpha 2 chain; COL3A1, Collagen Type III Alpha 1 Chain; ECM, extracellular matrix; FN1, fibronectin 1; Hic1, Hypermethylated in Cancer 1; LRRC15+, leucine-rich repeat-containing protein 15 positive; LWC_14d_neg, Hic1- cells from large wound center on day 14; LWC_14d_pos, Hic1+ cells from large wound center (1.5 cm in diameter) on day 14; LWP_14d_neg, Hic1- cells from large wound periphery on day 14; LWP_14d_pos, Hic1+ cells from

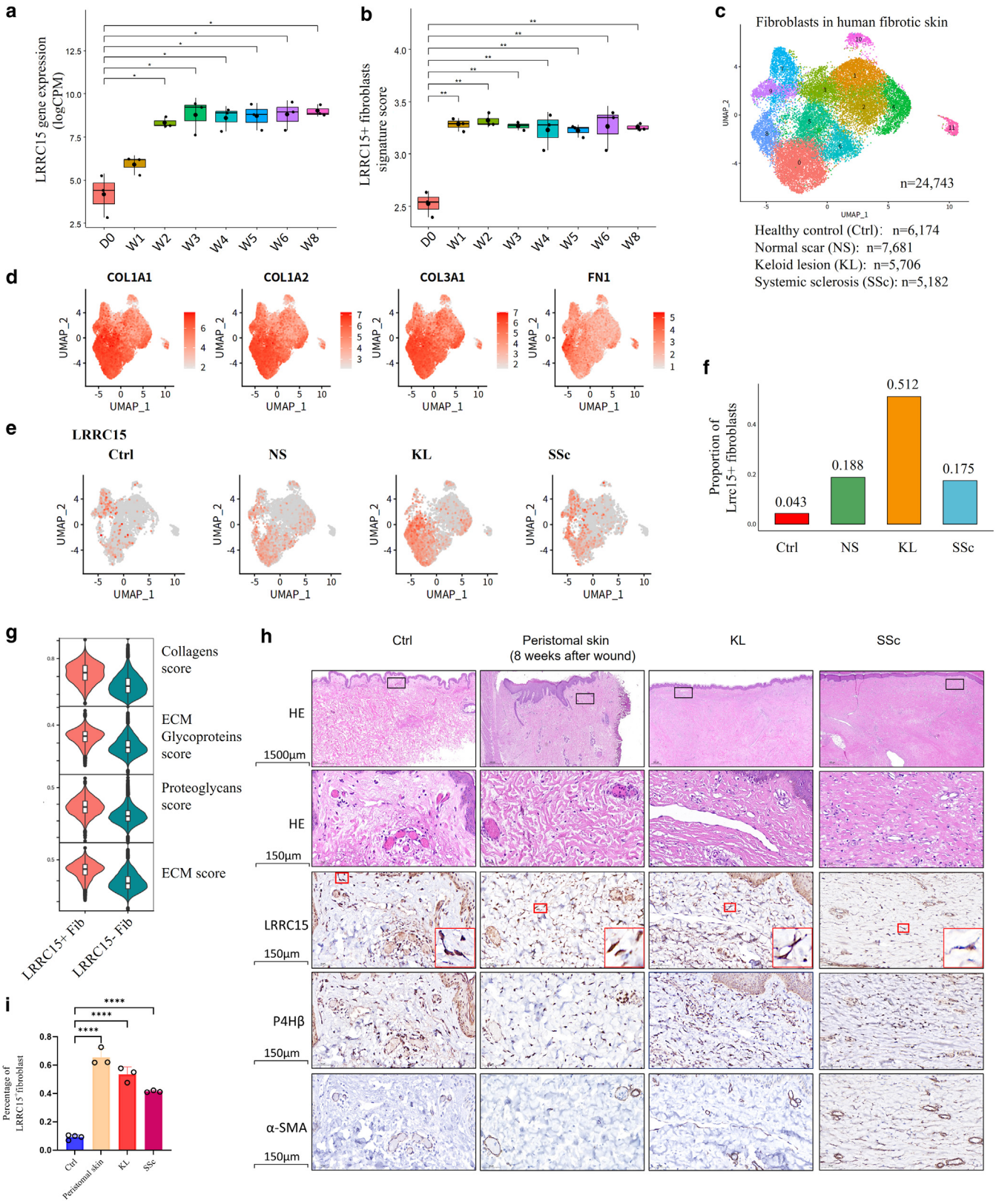

Figure 2. Expression of LRRC15+ fibroblasts in human wound healing and fibrotic diseases. (**a**) Bulk RNA-seq analysis of LRRC15 expression in human wound healing skin tissues at 0 days to 8 weeks, covering 8 time points. $*P < .05$. (**b**) Abundance of LRRC15+ fibroblast gene signature in human wound healing skin tissues at various time points. $**P < .01$. (**c**) UMAP plot of 24,743 fibroblasts from skin tissues of healthy Ctrl, NS, KL, and SSc. (**d**) UMAP plot color-coded by the expression of fibroblast marker genes, such as COL1A1, COL1A2, COL3A1, and FN1. (**e**) UMAP plot color-coded by the expression of LRRC15 in Ctrl, NS, KL, large wound periphery on day 14; LWP_18d_neg, Hic1- cells from large wound periphery on day 18; LWP_18d_pos, Hic1+ cells from large wound periphery on day 18; PWD12, post-wound day 12; PWD4, post-wound day 4; PWD7, post-wound day 7; SW_14d_pos, Hic1+ cells from small wound skin on day 14; SW_8d_pos, Hic1+ cells from small wound skin (8 mm in diameter) on day 8; UI_pos, Hic1+ cells from uninjured skin; UMAP, uniform manifold approximation and projection; UW, uninjured.

spleen in healthy murine tissues (Krishnamurty et al, 2022). We report here that LRRC15+ fibroblasts emerge during normal tissue repair processes. We detected LRRC15+ fibroblasts in relevant numbers in the healing phase of skin wounds, in scar formation, and in muscle repair. The involvement of LRRC15+ fibroblasts in the normal repair responses highlights the potential challenges of targeted depletion of LRRC15+ fibroblasts with negative effects on normal repair processes.

Using previously published single-cell RNA sequencing data from three perturbed human tissues (pancreas, intestine, and lung) (Buechler et al, 2021), we detected the highest numbers of LRRC15+ fibroblasts in pancreatic ductal adenocarcinoma, but also in NSCLC. In an atlas comprising 13 perturbed mouse tissues, most LRRC15+ fibroblasts were found in skin wounds, muscle injury, arthritis, and pancreatic ductal adenocarcinoma (Supplementary Figure S1a and b). Thus, we confirm the presence of LRRC15+ fibroblasts in tumors but also detect these cells in tissues with active remodeling in response to inflammation and injury.

To evaluate the capability of LRRC15+ fibroblasts to release extracellular matrix (ECM), we utilized a previously validated ECM score that incorporates the mRNA expression profiles of all collagens, glycoproteins, and proteoglycans. When compared to other fibroblast subtypes, LRRC15+ fibroblasts exhibited the highest collagens score, ECM glycoproteins score, proteoglycans score, and total ECM score in both human and mouse datasets Supplementary Figure S1c and d). These findings suggest that LRRC15+ fibroblasts might be implicated in ECM remodeling, a process crucial for normal wound healing and fibrotic tissue remodeling.

To provide further evidence for the role of LRRC15+ fibroblasts in normal and pathological repair processes, we analyzed changes in the frequency of LRRC15+ fibroblasts during wound healing in mice. We utilized single-cell RNA sequencing data from three wound healing datasets encompassing skin from normal mice before and 4, 7, and 12 days post-wounding (Guerrero-Juarez et al, 2019; Haensel et al, 2020; Vu et al, 2022). In uninjured murine skin, LRRC15+ fibroblasts comprised 5.4% of all fibroblasts, with increases to 23.3% on the fourth day after wounding with peaking at 71.8% after 12 days (Figure 1a–d). Simultaneously, we obtained consistent results in an additional dataset with a longer duration of wound healing (Abbasi et al, 2021). LRRC15+ fibroblasts also demonstrated higher ECM scores than other fibroblast populations (Figure 1e–i).

To confirm the relevance of these findings for repair processes in humans, we analyzed RNA sequencing datasets of human wound healing and tissue fibrosis. In human skin wounds with samples obtained at different time points after wounding (Ud-Din et al, 2019), we observed an upregulation of LRRC15 gene expression starting in the second week and persisting throughout the observation period of eight weeks (Figure 2a). Using the top 50 marker genes specific to LRRC15+ fibroblasts as a signature, we conducted gene enrichment analysis to reveal early and persistent upregulation of LRRC15+ fibroblast marker genes in wound healing of human skin (Figure 2b). We also observed increased numbers of LRRC15+ fibroblasts in different human fibrotic conditions. Comparison of healthy human skin (control), normal skin scars, keloid lesions as localized fibrosis of the skin (Deng et al, 2021), and systemic sclerosis as a disease with diffuse fibrotic remodeling of the skin and internal organs (Tabib et al, 2021). We noted that 4.3% of fibroblasts in healthy human skin express LRRC15+, whereas in normal skin scars, this proportion increased to 18.8%. In keloid lesions, 51.2% of fibroblasts were LRRC15-positive, and 17.5% in systemic sclerosis skin. As in the other datasets, LRRC15+ fibroblasts showed higher ECM scores than other fibroblast populations (Figure 2c–g).

To further validate the increase of LRRC15+ fibroblasts with an additional experimental approach, we quantified the number of LRRC15+ fibroblasts in normal skin, healing peristomal skin, keloids skin tissue, and systemic sclerosis skin by immunohistochemistry. We utilized prolyl-4-hydroxylase-β and α-smooth muscle actin as established markers for fibroblasts and myofibroblasts, respectively. LRRC15 expressing fibroblasts were strongly increased in healing peristomal skin, keloid skin tissue, and systemic sclerosis skin compared to normal skin. Virtually all LRRC15+ fibroblasts co-express prolyl-4-hydroxylase-β, whereas only a small subset of LRRC15+ fibroblasts express α-smooth muscle actin (Figure 2h and i).

In conclusion, we present evidence that LRRC15+ fibroblasts are not restricted to tumor tissues, but occur in significant numbers in murine and human conditions with high ECM turnover such as wound healing and fibrotic tissue remodeling, and may play an active role in ECM remodeling. Targeting of LRRC15 fibroblasts may thus not only offer the potential to target cancer-associated fibroblasts in cancer therapy (Krishnamurty et al, 2022) but also in fibrotic diseases. However, depletion of LRRC15 fibroblasts may pose a risk of impaired wound healing, for example during the post-tumor surgery recovery. Organ-specific or local targeting of LRRC15+ fibroblasts may minimize these side effects.

ETHICS STATEMENT

The ethics committee of Xiangya Hospital, Central South University, approved the protocol for human experimentation in this study, ensuring all

and SSc groups. (**f**) Fraction of LRRC15+ fibroblasts in each condition. (**g**) Violin plots illustrating scores for Collagens, ECM glycoproteins, proteoglycans, and ECM in human LRRC15+ and LRRC15- fibroblast clusters. (**h**) Representative histological images include H&E staining, as well as IHC images of LRRC15, the P4Hβ, and α-SMA staining (myofibroblast marker) in consecutive sections of skin tissues obtained from normal skin (Ctrl), peristomal skin from an intestinal stoma, KL, and SSc. LRRC15 is present in both the nuclei and cytoplasm of fibroblasts and LRRC15+ fibroblasts are found throughout the skin tissue. (**i**) Quantification of LRRC15+ fibroblast in skin tissue sections from 4 normal skin (Ctrl), 3 patients with peristomal skin, 3 patients with KL, and three patients with SSc. ****$P < .0001$. COL1A1, collagen type I alpha 1 chain; COL1A2, collagen type I alpha 2 chain; COL3A1, collagen type III alpha 1 chain; Ctrl, control; ECM, extracellular matrix; FN1, fibronectin 1; IHC, immunohistochemistry; KL, keloid lesions; LRRC15, leucine-rich repeat-containing protein 15; NS, normal scars; P4Hβ, prolyl 4-hydroxylase subunit β; RNA-seq, RNA sequencing; SSc, systemic sclerosis; UMAP, uniform manifold approximation and projection; α-SMA, α-smooth muscle actin.

procedures were conducted anonymously. Participants willingly joined the study and provided informed, written consent. The research adhered to the ethical guidelines outlined in the 1964 Helsinki Declaration, including subsequent amendments and equivalent standards. The Methods section is provided in the supplemental file.

DATA AVAILABILITY STATEMENT

All data used in this study are currently accessible via the public databases (https://fibroXplorer.com, GSE138669, GSE163973, GSE188432, GSE113854, GSE142471, GSE108677, GSE124161).

KEYWORDS

Cancer-associated fibroblasts; LRRC15+ fibroblasts; Wound healing

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CONFLICT OF INTEREST

JHWD has consultancy relationships with AbbVie, Active Biotech, Anamar, ARXX, AstraZeneca, Bayer Pharma, Boehringer Ingelheim, Celgene, Galapagos, Genentech, GSK, Inventiva, Janssen, Novartis, Pfizer, Roche and UCB. JHWD has received research funding from AbbVie, Anamar, Argenx, ARXX, BMS, Bayer Pharma, Boehringer Ingelheim, Cantargia, Celgene, CSL Behring, ExoTherapeutics, Galapagos, GSK, Inventiva, Kiniksa, Lassen, Novartis, Sanofi-Aventis, RedX, UCB. JHWD is the stock owner of 4D Science and Scientific Lead of FibroCure. None of the authors has a conflict of interest directly related to LRRC15.

ACKNOWLEDGMENTS

This research was funded by Grants (DI 1537/17-1, DI 1537/20-1, DI 1537/22-1, DI 1537/23-1) of the German Research Foundation, SFB TR221/ project number 324392634 (B04) of the German Research Foundation, a Career Support Award of Medicine of the Ernst Jung Foundation and an unrestricted research grant from the Hiller-Foundation. Honglin Zhu is the guarantor of the study.

AUTHOR CONTRIBUTIONS

Conceptualization: HZ, JHWD; Data Curation: HZ, MM, HL, JHWD; Formal Analysis: HZ, MM, HL, JHWD; Software: HZ, JHWD; Supervision: HZ, HL, JHWD; Validation: HZ, MM, HL, JHWD; Visualization: HZ, MM; Writing - Original Draft Preparation: HZ, HL, JHWD; Writing - Review and Editing: HZ, MM, JHWD

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SUPPLEMENTARY MATERIAL

Supplementary material is linked to the online version of the paper at www.jidonline.org, and at https://doi.org/10.1016/j.jid.2024.06.1279.

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Vu R, Jin S, Sun P, Haensel D, Nguyen QH, Dragan M, et al. Wound healing in aged skin exhibits systems-level alterations in cellular composition and cell-cell communication. Cell Rep 2022;40:111155.

Loss of ERBB2 and ERBB3 Receptors Impacts Epidermal Differentiation in Mice

Journal of Investigative Dermatology (2025) **145,** 204–208; doi:10.1016/j.jid.2024.06.1278

TO THE EDITOR

Antibody-based cancer therapy has achieved significant success, and mAbs targeting ERBB2 (trastuzumab, pertuzumab) and ERBB3 (patritumab, seribantumab, lumretuzumab) are used to treat various types of cancer (Propper et al, 2023). The ERBB2/3 heterodimer, the most active signaling dimer in the ERBB family (Schneider and Yarden, 2016), is often hyperactive in tumors, making it a promising target in cancer therapy. Although antibodies targeting EGFR and their toxicity are largely recognized (Lichtenberger et al, 2013), the side effects of targeting the related Y kinase receptors ERBB2 and ERBB3 remain to be determined (Braden and Anadkat, 2016; Hynes and Lane, 2005).

The development of anticancer drugs depends on preclinical research based on reliable animal models able to predict the efficacy and toxicity of candidate drugs or drug combinations. Therefore, we generated a skin-specific double knockout for both receptors

Abbreviation: K, keratin

Accepted manuscript published online 8 July 2024; corrected proof published online 31 July 2024

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SUPPLEMENTARY MATERIALS AND METHODS

Single-cell RNA sequencing data analysis

The integrated objects of perturbed-state human and mouse fibroblasts were downloaded from https://fibroXplorer.com (Buechler et al, 2021). For human fibrosis, single-cell RNA sequencing data were sourced from the Gene Expression Omnibus Series 138669, comprising skin samples from 12 systemic sclerosis patients and 10 healthy controls (Tabib et al, 2021). Additionally, human keloid and normal scar samples were gathered from GSE163973 (Deng et al, 2021). The data analysis was conducted using Seurat V4.4.0, and harmony was employed to integrate the two datasets. For the mouse wound healing, single-cell RNA sequencing data were obtained from three datasets: GSE188432 (young mouse in the unwounded state, and at 4- and 7-days post-wounding) (Vu et al, 2022), GSE113854 (post-wounding day 12) (Guerrero-Juarez et al, 2019), and GSE142471 (unwounded state and at 4 days post-wounding) (Haensel et al, 2020). Single-cell RNA sequencing data for the Hic1^{CreERT2}:Rosa^{tdTomato} mice were downloaded from GSE108677 (Abbasi et al, 2021). Reciprocal principal component analysis was used to integrate data from all samples. Uniform manifold approximation and projection were used for visualization. The main clusters were first annotated by SingleR (v1.4.0) (Aran et al, 2019), and fibroblast subpopulations were further annotated by marker genes (collagen type I alpha 1 chain, collagen type I alpha 2 chain, collagen type III alpha 1 chain, and fibronectin 1). Cells were scored using Seurat's AddModuleScore function, relying on the extracellular matrix gene sets (Naba et al, 2016).

Bulk RNA sequencing data analysis and leucine-rich repeat-containing protein 15 positive fibroblast abundance calculation

The human wound healing bulk RNA sequencing data were downloaded from GSE124161 (Ud-Din et al, 2019). RNA-sequencing count data were normalized using log-transformed counts per million. The scores for leucine-rich repeat-containing protein 15 positive fibroblasts in each sample were calculated using gene set variation analysis (Hänzelmann et al, 2013). This analysis utilized the top 50 marker genes of leucine-rich repeat-containing protein 15 positive fibroblasts and bulk RNA-sequencing data to derive the enrichment score.

Immunohistochemical staining

Skin tissues were collected from 4 normal controls, 3 peristomal skin of the intestinal stoma, 3 keloids, and 3 systemic sclerosis. All donors willingly participated in the study and provided written informed consent. The skin tissues were embedded in paraffin, and sections were stained with H&E for histological examination. For immunohistochemical staining, antigen retrieval was performed by pressure cooking (sections in EDTA, pH 8.0) for 8 minutes. Subsequently, the slides underwent a 10-minute incubation with 3% hydrogen peroxide, followed by exposure to 5% goat serum to block non-specific binding at 37 °C for 30 minutes. The slides were then immunostained with monoclonal antibodies against leucine-rich repeat-containing protein 15 (rabbit monoclonal, ab150376, Abcam, Cambridge, MA, USA), prolyl-4-hydroxylase-β (rabbit monoclonal, ab137110, Abcam, Cambridge, MA, USA), or α-smooth muscle actin (ZM-0003, Zsbio, Beijing, China) for 1 hour at 37 °C and 8 hours at 4 °C , with dilutions of 1:200, 1:1000, and 1:1000, respectively. After washing in Tris-buffered saline-tween, the slides were incubated with biotinylated goat anti-mouse antibody (Zsbio, Beijing, China) for 30 minutes at 37 °C. Staining was visualized using a diaminobenzidine solution kit (Zsbio, Beijing, China). Immunohistochemical images were scanned with the Pannoramic scanner and analyzed using Caseviewer software (3DHISTECH Ltd, Budapest, Hungary).

Statistics

All data normality was determined by the Shapiro-Wilks Normality Test. For continuously distributed data, between-group comparisons were performed using an unpaired *t*-test (if both 2 independent samples were normally distributed) or the Wilcoxon test (if at least one of the independent samples was not normally distributed). For the multiple *t*-tests, false discovery rate correction was applied to adjust the *P*-values. *P*-values below .05 are considered statistically significant. All analyses were conducted in R (v4.0.3) with the package MVN (v5.8), rstatix (v0.6.0).

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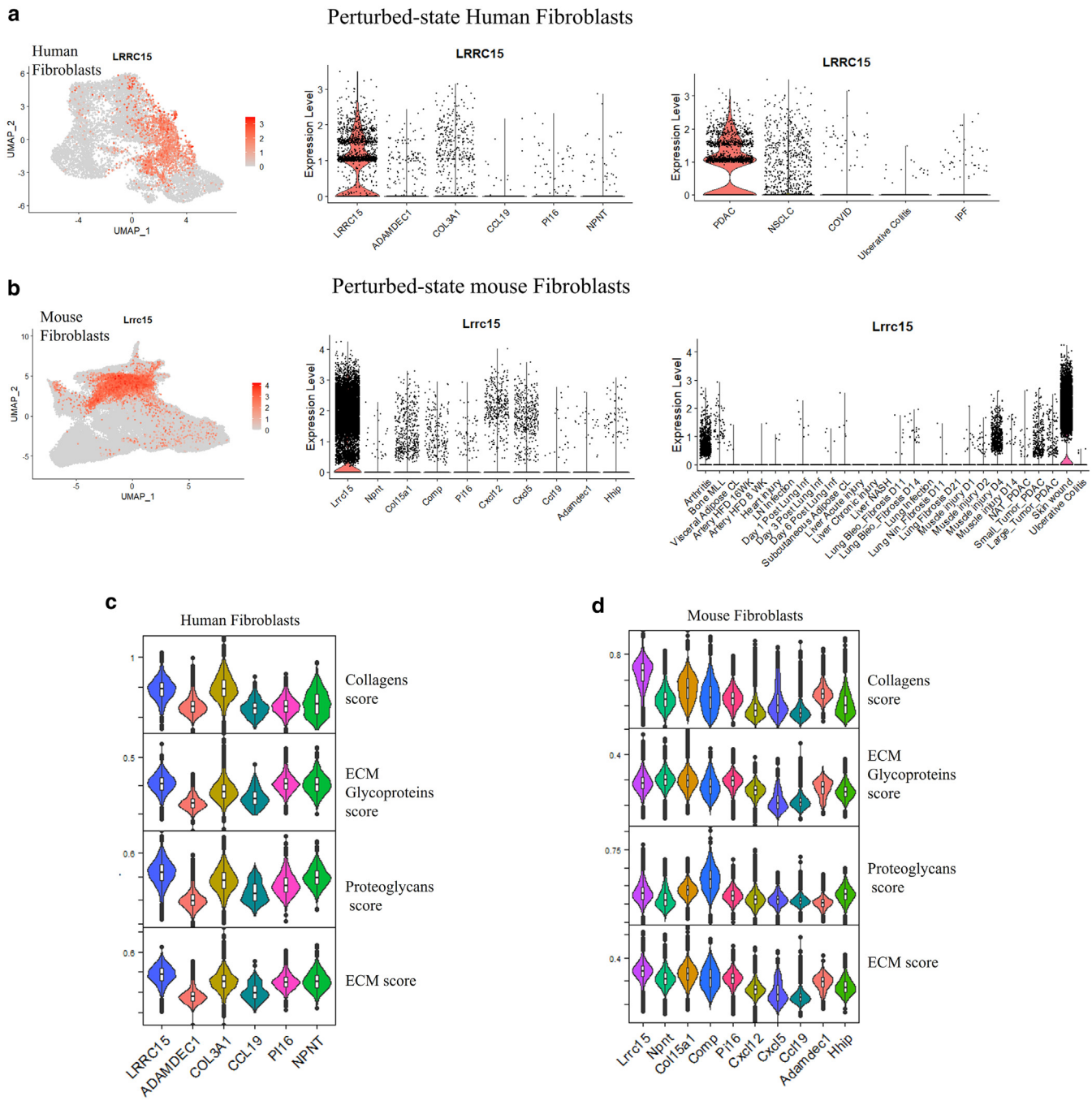

Supplementary Figure S1. LRRC15 expression in perturbed-state human and mouse fibroblasts. (**a**) scRNA-seq analysis of fibroblasts from three perturbed human tissues (pancreas, intestine, and lung) and five diseases (PDAC, NSCLC, COVID, Ulcerative Colitis, and IPF). UMAP plot of 21,262 fibroblasts colored by LRRC15 expression (left). Violin plot illustrating the distribution of LRRC15+ fibroblasts across the fibroblast clusters (middle) and perturbed human tissues (right). (**b**) scRNA-seq analysis of fibroblasts from 13 perturbed mouse tissues. UMAP plot of 99,596 fibroblasts colored by LRRC15 expression (left). Violin plot displaying the distribution of LRRC15+ fibroblasts across fibroblast clusters (middle) and perturbed mouse tissues (right). (**c**) Violin plots depicting scores for Collagens, ECM glycoproteins, proteoglycans, and total ECM in human fibroblast clusters. (**d**) Violin plots illustrating scores for Collagens, ECM glycoproteins, proteoglycans, and total ECM in mouse fibroblast clusters. COVID, coronavirus disease; ECM, extracellular matrix; IPF, idiopathic pulmonary fibrosis; LRRC15, leucine-rich repeat-containing protein 15; NSCLC, non-small cell lung cancer; PDAC, pancreatic ductal adenocarcinoma; scRNA-seq, single-cell RNA sequencing; UMAP, uniform manifold approximation and projection.